

Optimal Auctions

Jonathan Levin¹

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¹These slides are based on Paul Milgrom's.

Optimal Auctions

- What auction rules lead to the highest average prices?
- Difficulty: the set of auction rules is enormous, e.g.:
 - The auction is run in two stages. In stage one, bidders submit bids and each pays the average of its own bid and all lower bids.
 - The highest and lowest first stage bid are excluded. All other bidders are asked to bid again.
 - At the second stage, the item is awarded to a bidder with a probability proportional to the square of its second-stage bid. The winning bidder pays the average of its bid and the lowest second-stage bid.
- Can we really optimize revenue over *all* mechanisms?

Revelation Principle

- For any augmented Bayesian mechanism (Σ, ω, σ) , there is a corresponding direct mechanism (T, ω') for which truthful reporting is a Bayes-Nash equilibrium.
- The outcome function for the corresponding direct mechanism is given by:

$$\omega'(t) = \omega(\sigma_1(t_1), \dots, \sigma_N(t_N)).$$

Auction Revenues

- Suppose that
 - value of good to bidder i is $v_i(t_i)$
 - each v_i is increasing and differentiable.
 - types are independent, drawn from $U[0, 1]$.

Definition

An augmented mechanism (Σ, ω, σ) is **voluntary** if the maximal payoff $V_i(t_i)$ is non-negative everywhere.

- The expected revenue from an augmented mechanism is the expected sum of payments:

$$R(\Sigma, \omega, \sigma) \triangleq \mathbb{E} \left[\sum_{i=1}^N p_i^{\omega}(\sigma_1(t_1), \dots, \sigma_N(t_N)) \right].$$

Revenue maximization problem

- Allow randomized mechanisms, and let $x_i(t)$ denote the *probability* that i is awarded the good at equilibrium for the augmented mechanism.
- Then, we have the following constraints on feasible mechanisms:

$$\begin{aligned}x_i(\mathbf{t}) &\geq 0 \text{ for all } i \neq 0 \\ \sum_{i=0}^N x_i(\mathbf{t}) &\leq 1\end{aligned}$$

- Consider the problem

$$\max_{(\Sigma, \omega), \sigma \in NE} R(\Sigma, \omega, \sigma).$$

- Performance: $x_i(t_1, \dots, t_N) \triangleq x_i^\omega(\sigma_1(t_1), \dots, \sigma_N(t_N))$.

Marginal revenue

- Simple case... assume there is a *single* bidder with value $v(t)$, where v is increasing and t is drawn from $U[0, 1]$.
- If the seller fixes a price $v(s)$ it sells whenever $t \geq s$, or with probability $1 - s$.
 - Can view $1 - s$ as the “quantity” sold
 - Expected *total revenue* is $(1 - s)v(s)$.
 - *Marginal revenue* is $d\text{Rev}/dQ$:

$$MR(s) = v(s) - (1 - s)v'(s) = \frac{d((1 - s)v(s))}{d(1 - s)}$$

Theorem

Consider any augmented mechanism $(N, \Sigma, \omega, \sigma^*)$ for which σ^* is a BNE, $x_j(t)$ is the probability that j is allocated the good at type profile $\mathbf{t}$, and $V_j(0)$ is his expected payoff with type equal to zero. Then:

$$R(\Sigma, \omega, \sigma^*) = \int_0^1 \cdots \int_0^1 \sum_{i=1}^N x_i(s_1, \dots, s_N) MR_i(\mathbf{s}) ds_1 \dots ds_N - \sum_{i=1}^N V_i(0).$$

- Note that:

- Expected revenue is expressed as a function of the allocation x but not the payment function p .
- The expression is *linear* in the x_i 's and depends critically on the *marginal revenues*.

Derivation of revenue formula, I

- Consider mechanism (N, Σ, ω) with equilibrium profile σ^* .
- Consider a given bidder i . Its equilibrium payoff is:

$$\begin{aligned} V_i(t_i) &= \max_{\sigma_i \in \Sigma_i} \mathbb{E}_{t_{-i}} [x_i(\sigma_i, \sigma_{-i}^*(t_{-i})) v_i(t_i) - p_i(\sigma_i, \sigma_{-i}^*(t_{-i}))] \\ &= \max_{\sigma_i \in \Sigma_i} \mathbb{E}_{t_{-i}} [x_i(\sigma_i, \sigma_{-i}^*(t_{-i}))] v_i(t_i) - \mathbb{E}_{t_{-i}} [p(\sigma_i, \sigma_{-i}^*(t_{-i}))] \end{aligned}$$

- The derivative of this objective wrt t_i is:

$$\mathbb{E}_{t_{-i}} [x_i(\sigma_i, \sigma_{-i}^*(t_{-i}))] v_i'(t_i)$$

- By the envelope theorem this expression gives $V_i'(t)$...

Derivation of revenue formula, II

- Applying Myerson's Lemma, we have *for any* i ,

$$\begin{aligned} V_i(t_i) - V_i(0) &= \int_0^{t_i} \mathbb{E}_{s_{-i}} [x_i(s_i, s_{-i})] v'_i(s_i) ds_i \\ &= \int_0^{t_i} \left(\int_0^1 \cdots \int_0^1 x_i(s_1, \dots, s_N) ds_{-i} \right) v'_i(s_i) ds_i \end{aligned}$$

- So the bidder's *average* payoff across all types is:

$$\mathbb{E}_{t_i} [V_i(t_i)] - V_i(0) = \int_0^1 [V_i(t_i) - V_i(0)] dt_i$$

Derivation of revenue formula, III

- Using integration by parts we obtain

$$\begin{aligned} & \mathbb{E}_{t_i} [V_i(t_i)] - V_i(0) \\ &= \int_0^1 \left[\int_0^{t_i} \mathbb{E}_{s_{-i}} [x_i(s_i, s_{-i})] v'_i(s_i) ds_i \right] dt_i \\ &= \int_0^1 \mathbb{E}_{s_{-i}} [x_i(s_i, s_{-i})] v'_i(s_i) ds_i - \int_0^1 t_i \mathbb{E}_{s_{-i}} [x_i(t_i, s_{-i})] v'_i(t_i) dt_i \\ &= \int_0^1 (1 - s_i) \mathbb{E}_{s_{-i}} [x_i(s_i, s_{-i})] v'_i(s_i) ds_i \\ &= \int_0^1 \cdots \int_0^1 (1 - s_i) x_i(s_1, \dots, s_N) v'_i(s_i) ds_1 \dots ds_N \end{aligned}$$

Derivation of revenue formula, IV

- Total realized surplus is

$$\sum_{i=1}^N x_i(t_1, \dots, t_N) v_i(t_i) = x(\mathbf{t}) \cdot v(\mathbf{t})$$

- Total expected revenue is therefore

$$R(\Sigma, \omega, \sigma)$$

$$\begin{aligned} &= \mathbb{E}_{\mathbf{t}} [x(\mathbf{t}) \cdot v(\mathbf{t})] - \sum_{i=1}^N \mathbb{E}_{t_i} [V_i(t_i)] \\ &= \int_0^1 \cdots \int_0^1 x_i(s_1, \dots, s_N) [v_i - (1 - s_i) v_i'(s_i)] ds_1 \dots ds_N - \sum_{i=1}^N V_i(0) \\ &= \int_0^1 \cdots \int_0^1 x_i(s_1, \dots, s_N) MR_i(s_i) ds_1 \dots ds_N - \sum_{i=1}^N V_i(0) \end{aligned}$$

Theorem

Suppose that MR_i is increasing for all i . Then an augmented voluntary mechanism is expected revenue maximizing if and only if (i) $V_i(0) = 0$ for all i , (ii) bidder i is allocated the good exactly when $MR_i(t_i) > \max \{0, \max_{j \neq i} MR_j(t_j)\}$. Furthermore at least one such augmented mechanism exists, with expected revenue of:

$$\mathbb{E}_t [\max \{0, MR_1(t_1), \dots, MR_N(t_N)\}].$$

- From the prior theorem we have

$$\begin{aligned} R(\Sigma, \omega, \sigma) &= \int_0^1 \cdots \int_0^1 x_i(s_1, \dots, s_N) MR_i(s_i) ds_1 \dots ds_N - \sum_{i=1}^N V_i(0) \\ &\leq \int_0^1 \cdots \int_0^1 \max \{0, MR_1(s_1), \dots, MR_N(s_N)\} ds_1 \dots ds_N \end{aligned}$$

- The “0” in the max expression can be viewed as a “reserve price,” that is a condition under which the item is not sold.
- We show on the next slide that the revenue bound can be achieved using a dominant strategy mechanism.

Proof, II: an optimal auction

- Ask bidders to report their types. Assign the item to the bidder with the highest marginal revenue, provided it exceeds zero.
- Payments are

$$p_i(\mathbf{t}) = x_i(\mathbf{t}) \cdot v_i \left(MR_i^{-1} \left(\max \left\{ 0, \max_{j \neq i} MR_j(t_j) \right\} \right) \right).$$

- So bidder i pays only if she gets the item, and in that event she pays the value corresponding to the lowest type she could have reported and still been awarded the item.
- It's dominant to report one's true type and $V_i(0) = 0$ for all i .

Example

- An optimal auction may distort the allocation away from efficiency to increase revenue.
 - Bidder 1's value $v_1(t_1) = t_1$, so value is $U[0, 1]$.
 - Bidder 2's value $v_2(t_2) = 2t_2$, so value is $U[0, 2]$.
- Marginal revenues:

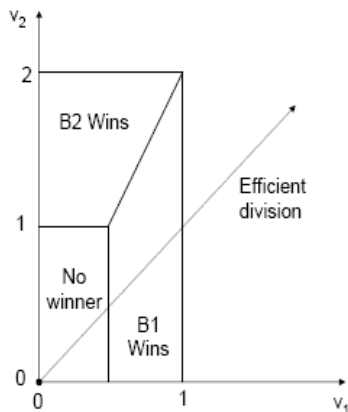
$$MR_1(t_1) = t_1 - (1 - t_1) = 2t_1 - 1 = 2v_1 - 1$$

$$MR_2(t_2) = 2t_2 - 2(1 - t_2) = 4t_2 - 2 = 2v_2 - 2.$$

- Bidder 1 may win despite having lower value, and auction may not be awarded despite both bidders having positive value.

Example, cont.

- Allocation in the optimal auction



Relation to Monopoly Theory (Bulow-Roberts)

- The problem is analogous to standard multi-market monopoly problem.
 - Each bidder is a separate “market” in which the good can be sold.
 - Quantity is analogous to probability of winning – probabilities must sum to one like a quantity constraint.
 - Monopolist can price discriminate, setting different prices in different markets.
 - Allocating the probability of winning to individual markets is analogous to allocating quantities across markets.
- Solution in both cases: sell marginal unit where marginal revenue is highest, provided it is positive!

Corollary

Suppose valuation functions are identical $v_1 = \dots = v_N = v$ and $v(s)$, $MR(s) = v(s) - (1-s)v'(s)$ are increasing and take positive and negative values. A voluntary auction maximizes expected revenue iff at equilibrium, $V(0) = 0$, and the auction is assigned to the high value bidder provided its value exceeds $v(r)$, where r satisfies $MR(r) = 0$.

Corollary

Suppose the valuation functions are identical and $v(s)$, $MR(s)$ are increasing. Then the following auctions are optimal:

- *A second price auction with minimum bid $v(r)$.*
- *A first price auction with minimum bid $v(r)$.*
- *An ascending auction with minimum bid $v(r)$.*
- *An all-pay auction with minimum bid $v(r)r^{N-1}$.*

Reserve Price Example

- Suppose N bidders and $v(t) = t$ for all bidders.
- From above, $MR(t) = 2t - 1$.
- The optimal reserve price is $v(r) = 1/2$, i.e. from $MR(r) = 0$.
- In a second price auction, each bidder bids its value, but doesn't bid if $t < 1/2$.
- In a first price auction, each bidder places no bid if $t < 1/2$ and otherwise bids $\beta(t) = \left(\frac{N-1}{N} + \frac{(2t)^{-N}}{N} \right) t$.

Theorem

Suppose that the marginal revenue function is increasing and that $v_i(0) = 0$ for all i . Then, adding a single buyer to an “otherwise optimal auction but with zero reserve” yields more expected revenue than setting the reserve optimally, that is,

$$\begin{aligned} \mathbb{E} [\max \{MR_1(t_1), \dots, MR_N(t_N), MR_{N+1}(t_{N+1})\}] \\ \geq \mathbb{E} [\max \{MR_1(t_1), \dots, MR_N(t_N), 0\}]. \end{aligned}$$

Lemma

$$\mathbb{E}[MR_i(t_i)] = v_i(0)$$

Proof.

$$TR_i(s) = (1 - s) v_i(s)$$

$$MR_i(s) = \frac{d}{d(1-s)} TR_i(s) = -\frac{d}{ds} TR_i(s)$$

so therefore

$$\mathbb{E}[MR_i(s)] = \int_0^1 MR_i(s) ds = -[TR_i(1) - TR_i(0)] = v_i(0).$$

Lemma

Given any random variable X and any real number α ,

$$\mathbb{E} [\max\{X, \alpha\}] \geq \mathbb{E} [X] \text{ and}$$

$$\mathbb{E} [\max\{X, \alpha\}] \geq \alpha, \text{ so}$$

$$\mathbb{E} [\max\{X, \alpha\}] \geq \max\{\mathbb{E} [X], \alpha\}.$$

Proof. This follows from Jensen's inequality.

Bulow-Klemperer Proof

- Applying the two lemmata, with $X = MR_{N+1}(t_{N+1})$,

$$\begin{aligned} & \mathbb{E} [\text{Revenue, } N + 1 \text{ bidders, no reserve}] \\ &= \mathbb{E}_{t_1, \dots, t_N, t_{N+1}} [\max \{ MR_1(t_1), \dots, MR_N(t_N), MR_{N+1}(t_{N+1}) \}] \\ &= \mathbb{E}_{t_1, \dots, t_N} [\mathbb{E}_{t_{N+1}} [\max \{ MR_1(t_1), \dots, MR_N(t_N), MR_{N+1}(t_{N+1}) \}]] \\ &\geq \mathbb{E}_t [\max \{ MR_1(t_1), \dots, MR_N(t_N), 0 \}] \\ &= \mathbb{E} [\text{Revenue, } N \text{ bidders, optimal auction}]. \end{aligned}$$

Using the RET...

- Bulow-Klemperer theorem is one example of how to use the RET to derive new results in auction theory.
- Many other results also rely on RET arguments
 - McAfee-McMillan (1992) weak cartels theorem
 - Weber (1983) analysis of sequential auctions
 - Bulow-Klemperer (1994) analysis of dutch auctions
 - Bulow-Klemperer (1999) analysis of war of attrition.
- Milgrom's chapter five contains many examples.